IGCSE 0607 Practice Questions — E2.2 — Combined Questions & Markscheme

Algebraic manipulation

Total Questions: 21 — **Total Marks:** 120 Marks

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- **E2.2** Algebraic manipulation
  → **Paper 2 (Non-Calculator)** *(16 Qs)* · **Paper 4 (Calculator)** *(5 Qs)*

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1 **(a)** Expand and simplify.

$4(2x - 3y) - 3(x - 5y)$

..... **[2]**

(b) Factorise completely.

$14a^2b - 21ab^2$

..... **[2]**

2 **(a)** Expand and simplify.

$3(2x - 5) - 2(x + 4)$

..... **[2]**

(b) Factorise completely.

$8x^2y - 12xy^2$

..... **[2]**

3 **(a)** Expand and simplify

$$3(2x - 5) - 4(x - 2)$$

..... **[2]**

(b) Factorise fully

$$12x^2y - 18xy^2$$

..... **[2]**

4 **(a)** Expand and simplify.

$3(2x - 5) - 4(x - 3)$

..... **[2]**

(b) Factorise fully.

$14x^2y + 21xy^2$

..... **[2]**

5 **(a)** Expand and simplify.

$(2x - 3)(x + 5)$

..... **[2]**

(b) Factorise.

$ac - 3bc + 2ad - 6bd$

..... **[2]**

(c) Factorise.

$16x^2 - 25y^2$

..... **[1]**

6 **(a)** Factorise.

$x^2 - 8x + 15$

..... **[2]**

(b) Factorise.

$2x^2 - 5x - 12$

..... **[2]**

(c) Factorise.

$ax - 3a + 2bx - 6b$

..... **[2]**

7 **(a)** Expand and simplify.

$(2x - 5)(3x + 1)$

..... **[2]**

(b) Factorise.

$x^2 - 7x - 18$

..... **[2]**

(c) Factorise completely.

$18x^2 - 50y^2$

..... **[3]**

8 **(a)** Factorise completely.

$18x^2 - 50y^2$

..... **[3]**

(b) Factorise completely.

$3x^3 + 10x^2 - 8x$

..... **[3]**

9 **(a)** Factorise

$$6ax - 2ay - 3bx + by$$

..... **[2]**

(b) Factorise completely.

(i)

$$50x^2 - 8y^2$$

..... **[3]**

(ii)

$$2x^3 + 5x^2 - 12x$$

..... **[3]**

10 **(a)** Expand and simplify

$$(x-3)(2x+1)(x+4)$$

..... **[3]**

(b) Factorise

$$2x^2-7x-15$$

..... **[2]**

11 **(a)** Expand and simplify.

$(x-4)(2x+3)(x+1)$

..... **[3]**

(b) Factorise completely.

$6x^2 - 11x - 10$

..... **[2]**

(c) Factorise completely.

$x^3 + 3x^2 - 4x - 12$

..... **[3]**

12 **(a)** Expand and simplify.

$(x-2)(x+3)(2x-1)$

..... **[3]**

(b) Factorise.

$6x^2 - 11x - 10$

..... **[2]**

(c) Factorise completely.

$2x^3 - 18x$

..... **[3]**

13 **(a)** Expand and simplify

$$4(3x - 2) - 2(x - 5)$$

..... **[2]**

(b) Factorise fully

$$15a^2b - 10ab^2$$

..... **[2]**

14 **(a)** Expand and simplify

$$(x-4)(2x+3)(x+2)$$

..... **[3]**

(b) Factorise

$$16x^2 - 49y^2$$

..... **[2]**

(c) Factorise

$$3x^2 - 10x - 8$$

..... **[2]**

15 **(a)** Factorise

$$cx - 5dx + 3cy - 15dy$$

..... **[2]**

(b) Factorise fully

$$2x^3 - 18x$$

..... **[2]**

(c) Factorise fully

$$12x^3 - 14x^2 - 10x$$

..... **[3]**

16 **(a)** Expand and simplify.

$3(2x - 1) - 2(x + 4)$

..... **[2]**

(b) Factorise fully.

$15x^2y - 20xy^2$

..... **[2]**

17 **(a)** Expand and simplify.

$(2x - 3)(x + 5)$

..... **[2]**

(b) Factorise.

$x^2 - 7x - 18$

..... **[2]**

(c) Factorise completely.

$4x^2 - 25y^2$

..... **[2]**

18 **(a)** Expand and simplify.

$$(x-2)(x+3)(2x-1)$$

..... **[3]**

(b) Factorise completely.

(i)

$$6x^2 - 11x - 10$$

..... **[2]**

(ii)

$$ax - ay + 3cx - 3cy$$

..... **[2]**

19 **(a)** Simplify.

$3a - 5b + 2a + 8b$

..... **[2]**

(b) Factorise completely.

$12x^2y - 18xy^2$

..... **[2]**

20 **(a)** Expand and simplify.

$(2x - 3)(x + 5)$

..... **[2]**

(b) Factorise.

$y^2 - 16x^2$

..... **[1]**

(c) Factorise.

$p^2 - 4pq - 5q^2$

..... **[2]**

21 **(a)** Expand and simplify.

$(x-4)(2x+1)(x+3)$

..... **[3]**

(b) Factorise completely.

$2ax - 3by + 2ay - 3bx$

..... **[2]**

(c) Factorise.

$6x^2 + 7x - 20$

..... **[2]**

1 **(a)** $8x - 12y - 3x + 15y$ **[M1]**

$5x + 3y$ **[A1]**

(b) $7ab$ extracted as a common factor **[M1]**

$7ab(2a - 3b)$ **[A1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** $6x - 15 - 2x - 8$ **[M1]**

$4x - 23$ **[A1]**

(b) $4xy(2x - 3y)$ or $4x(2xy - 3y^2)$ or $4y(2x^2 - 3xy)$ or $xy(8x - 12y)$ **[M1]**

$4xy(2x - 3y)$ **[A1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** Correct expansion: $6x - 15 - 4x + 8$ **[M1]**

$2x - 7$ **[A1]**

(b) $6xy(2x - 3y)$ **[B2]**

Correct partial factorisation e.g., $6x(2xy - 3y^2)$ or $2xy(6x - 9y)$ **[M1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** Expansion to give $6x - 15$ and $-4x + 12$ **[M1]**

$2x - 3$ **[A1]**

(b) $7xy(2x + 3y)$ **[B2]**

Partial factorisation, e.g. $7x(2xy + 3y^2)$ or $xy(14x + 21y)$ **[B1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** $2x^2 + 10x - 3x - 15$ **[M1]**

$2x^2 + 7x - 15$ **[A1]**

(b) $c(a - 3b) + 2d(a - 3b)$ or $a(c + 2d) - 3b(c + 2d)$ **[M1]**

$(a - 3b)(c + 2d)$ **[A1]**

(c) $(4x - 5y)(4x + 5y)$ **[B1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** $(x + a)(x + b)$ where $ab = 15$ or $a + b = -8$ **[M1]**

$(x - 3)(x - 5)$ **[A1]**

(b) $(2x + a)(x + b)$ where $ab = -12$ or $2b + a = -5$ **[M1]**

$(2x+3)(x-4)$ **[A1]**

(c) $a(x-3)+2b(x-3)$ or $x(a+2b)-3(a+2b)$ **[M1]**

$(a+2b)(x-3)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

7 **(a)** $6x^2+2x-15x-5$ (at least 3 correct terms) **[M1]**

$6x^2-13x-5$ **[A1]**

(b) $(x+a)(x+b)$ where $ab=-18$ or $a+b=-7$ **[M1]**

$(x-9)(x+2)$ **[A1]**

(c) $2(9x^2-25y^2)$ **[M1]**

$(3x-5y)(3x+5y)$ seen **[M1]**

$2(3x-5y)(3x+5y)$ **[A1]**

[Back to Question] *[Total Marks: 7]*

8 **(a)** Taking out common factor to reach $2(9x^2-25y^2)$ **[M1]**

Factorising difference of squares to reach $(3x-5y)(3x+5y)$ **[M1]**

$2(3x-5y)(3x+5y)$ **[A1]**

(b) Taking out common factor x to reach $x(3x^2+10x-8)$ **[M1]**

Factorising quadratic to reach $(3x-2)(x+4)$ **[M1]**

$x(3x-2)(x+4)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

9 **(a)** $(2a-b)(3x-y)$ **[B2]**

Correct partial factorisation e.g., $2a(3x-y)-b(3x-y)$ or $3x(2a-b)-y(2a-b)$ **[M1]**

(b) **(i)** Extracts common factor of 2: $2(25x^2-4y^2)$ **[M1]**

Factorises difference of two squares: $(5x-2y)(5x+2y)$ **[M1]**

$2(5x-2y)(5x+2y)$ **[A1]**

(ii) Extracts common factor of x: $x(2x^2+5x-12)$ **[M1]**

Factorises quadratic: $(2x-3)(x+4)$ **[M1]**

$x(2x-3)(x+4)$ **[A1]**

[Back to Question] *[Total Marks: 8]*

10 **(a)** Correct expansion of any two brackets, e.g., $(2x^2-5x-3)$ or $(2x^2+9x+4)$ or (x^2+x-12) **[M1]**

Correct expansion of their quadratic with the third bracket, producing at least 5 terms, e.g., $2x^3+8x^2-5x^2-20x-3x-12$ **[M1]**

$2x^3 + 3x^2 - 23x - 12$ **[A1]**

(b) $(2x + 3)(x - 5)$ **[B2]**

Accept $(2x + a)(x + b)$ where $ab = -15$ or $2b + a = -7$ **[M1]**

[Back to Question] *[Total Marks: 5]*

11 **(a)** Expands any two brackets correctly, e.g., $(2x^2 - 5x - 12)(x + 1)$ or $(x - 4)(2x^2 + 5x + 3)$ or $(x^2 - 3x - 4)(2x + 3)$ **[M1]**

Multiplies their resulting quadratic by the third bracket, allowing one error **[M1]**

$2x^3 - 3x^2 - 17x - 12$ **[A1]**

(b) $(3x + a)(2x + b)$ or $(6x + c)(x + d)$ where $ab = -10$ or $cd = -10$ **[M1]**

$(3x + 2)(2x - 5)$ **[A1]**

(c) $x^2(x + 3) - 4(x + 3)$ or $x(x^2 - 4) + 3(x^2 - 4)$ **[M1]**

$(x^2 - 4)(x + 3)$ **[M1]**

$(x - 2)(x + 2)(x + 3)$ **[A1]**

[Back to Question] *[Total Marks: 8]*

12 **(a)** $(x^2 + x - 6)(2x - 1)$ or $(x - 2)(2x^2 + 5x - 3)$ or $(x + 3)(2x^2 - 5x + 2)$ **[M1]**

$2x^3 - x^2 + 2x^2 - x - 12x + 6$ (allow one error in expansion) **[M1]**

$2x^3 + x^2 - 13x + 6$ **[A1]**

(b) $(3x \pm a)(2x \pm b)$ where $ab = 10$ or $6x^2 - 15x + 4x - 10$ **[M1]**

$(3x + 2)(2x - 5)$ **[A1]**

(c) $2x(x^2 - 9)$ **[M1]**

$(x - 3)(x + 3)$ seen **[M1]**

$2x(x - 3)(x + 3)$ **[A1]**

[Back to Question] *[Total Marks: 8]*

13 **(a)** $12x - 8 - 2x + 10$ **[M1]**

$10x + 2$ **[A1]**

(b) $5ab(3a - 2b)$ **[B2]**

If not B2, then B1 for $5a(3ab - 2b^2)$ or $5b(3a^2 - 2ab)$ or $ab(15a - 10b)$ **[B1]**

[Back to Question] *[Total Marks: 4]*

14 **(a)** $(x - 4)(2x^2 + 7x + 6)$ or $(2x^2 - 5x - 12)(x + 2)$ or $(x^2 - 2x - 8)(2x + 3)$ **[M1]**

$2x^3 + 7x^2 + 6x - 8x^2 - 28x - 24$ (allow one error in expansion) **[M1]**

$2x^3 - x^2 - 22x - 24$ **[A1]**

(b) $(4x - 7y)(4x + 7y)$ **[B2]**

If not B2, then B1 for $(4x - ky)(4x + ky)$ or $(kx - 7y)(kx + 7y)$ where $k \neq 0$ **[B1]**

(c) $(3x + 2)(x - 4)$ **[B2]**

If not B2, then B1 for $(3x + a)(x + b)$ where $ab = -8$ or $3b + a = -10$ **[B1]**

[Back to Question] *[Total Marks: 7]*

15 **(a)** $x(c - 5d) + 3y(c - 5d)$ or $c(x + 3y) - 5d(x + 3y)$ **[M1]**

$(x + 3y)(c - 5d)$ **[A1]**

(b) $2x(x^2 - 9)$ **[M1]**

$2x(x - 3)(x + 3)$ **[A1]**

(c) $2x(6x^2 - 7x - 5)$ **[M1]**

$2x(3x - 5)(2x + 1)$ **[A2]**

If not A2, then A1 for $(3x - 5)(2x + 1)$ or $2x(3x + 5)(2x - 1)$ **[A1]**

[Back to Question] *[Total Marks: 7]*

16 **(a)** Expansion to $6x - 3 - 2x - 8$ **[M1]**

$4x - 11$ **[A1]**

(b) Extracting a partial common factor, e.g. $5xy(3x - 4y)$ or $x(15xy - 20y^2)$ or $5(3x^2y - 4xy^2)$ **[M1]**

$5xy(3x - 4y)$ **[A1]**

[Back to Question] *[Total Marks: 4]*

17 **(a)** Four terms with at least three correct from $2x^2 + 10x - 3x - 15$ **[M1]**

$2x^2 + 7x - 15$ **[A1]**

(b) $(x + a)(x + b)$ where $ab = -18$ or $a + b = -7$ **[M1]**

$(x - 9)(x + 2)$ **[A1]**

(c) $(2x + a)(2x - a)$ or $(cx + 5y)(cx - 5y)$ **[M1]**

$(2x - 5y)(2x + 5y)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

18 **(a)** Correct expansion of two brackets, e.g. $(x^2 + x - 6)(2x - 1)$ or $(x - 2)(2x^2 + 5x - 3)$ **[M1]**

Multiplication of quadratic by third bracket yielding 6 terms, e.g. $2x^3 - x^2 + 2x^2 - x - 12x + 6$ **[M1]**

$2x^3 + x^2 - 13x + 6$ **[A1]**

(b) **(i)** $(3x + a)(2x + b)$ where $ab = -10$ or $2a + 3b = -11$ **[M1]**

$(3x+2)(2x-5)$ **[A1]**

(ii) $a(x-y)+3c(x-y)$ or $x(a+3c)-y(a+3c)$ **[M1]**

$(a+3c)(x-y)$ **[A1]**

[Back to Question] *[Total Marks: 7]*

19 **(a)** $5a$ or $+3b$ **[B1]**

$5a+3b$ **[A1]**

(b) $6xy(2x-3y)$ **[B2]**

If B0, allow [M1] for a correct partial factorisation, e.g. $6x(2xy-3y^2)$ or $3xy(4x-6y)$

[Back to Question] *[Total Marks: 4]*

20 **(a)** $2x^2+10x-3x-15$ **[M1]**

$2x^2+7x-15$ **[A1]**

(b) $(y-4x)(y+4x)$ **[B1]**

(c) $(p+q)(p-5q)$ **[B2]**

If B0, allow [M1] for $(p+aq)(p+bq)$ where $ab=-5$ or $a+b=-4$

[Back to Question] *[Total Marks: 5]*

21 **(a)** Expands any two brackets, e.g. $(2x^2-7x-4)(x+3)$ **[M1]**

Multiplies their quadratic by the third bracket, e.g. $2x^3-7x^2-4x+6x^2-21x-12$ **[M1]**

$2x^3-x^2-25x-12$ **[A1]**

(b) $2a(x+y)-3b(y+x)$ or $x(2a-3b)+y(2a-3b)$ **[M1]**

$(x+y)(2a-3b)$ **[A1]**

(c) $(2x+5)(3x-4)$ **[B2]**

If B0, allow [M1] for $(2x+a)(3x+b)$ where $ab=-20$ or $3a+2b=7$

[Back to Question] *[Total Marks: 7]*